

1. You are given a sorted integer array `arr` containing 1 and prime numbers, where all the integers of `arr` are unique. You are also given an integer `k`. For every `i` and `j` where $0 \leq i < j < \text{arr.length}$, we consider the fraction $\text{arr}[i] / \text{arr}[j]$. Return the `k`th smallest fraction considered. Return your answer as an array of integers of size 2, where `answer[0] == arr[i]` and `answer[1] == arr[j]`.

Example 1:

Input: `arr = [1,2,3,5]`, `k = 3`

Output: `[2,5]`

Explanation: The fractions to be considered in sorted order are:

$1/5$, $1/3$, $2/5$, $1/2$, $3/5$, and $2/3$.

The third fraction is $2/5$.

Example 2:

Input: `arr = [1,7]`, `k = 1`

Output: `[1,7]`

2. You are given an integer `n`. We reorder the digits in any order (including the original order) such that the leading digit is not zero. Return `true` if and only if we can do this so that the resulting number is a power of two.

Example 1:

Input: `n = 1`

Output: `true`

Example 2:

Input: `n = 10`

Output: `false`

3. A matrix diagonal is a diagonal line of cells starting from some cell in either the topmost row or leftmost column and going in the bottom-right direction until reaching the matrix's end. For example, the matrix diagonal starting from `mat[2][0]`, where `mat` is a 6 x 3 matrix, includes cells `mat[2][0]`, `mat[3][1]`, and `mat[4][2]`. Given an `m x n` matrix `mat` of integers, sort each matrix diagonal in ascending order and return the resulting matrix.

Example 1:

Input: `mat = [[3,3,1,1],[2,2,1,2],[1,1,1,2]]`

Output: `[[1,1,1,1],[1,2,2,2],[1,2,3,3]]`

Example 2:

Input: `mat =`

`[[11,25,66,1,69,7],[23,55,17,45,15,52],[75,31,36,44,58,8],[22,27,33,25,68,4],[84,28,14,11,5,50]]`

Output: `[[5,17,4,1,52,7],[11,11,25,45,8,69],[14,23,25,44,58,15],[22,27,31,36,50,66],[84,28,75,33,55,68]]`

4. The DNA sequence is composed of a series of nucleotides abbreviated as 'A', 'C', 'G', and 'T'. For example, "ACGAATTCCG" is a DNA sequence. When studying DNA, it is useful to identify repeated sequences within the DNA. Given a string `s` that represents a DNA sequence, return all the 10-letter-long sequences (substrings) that occur more than once in a DNA molecule. You may return the answer in any order.

Example 1:

Input: `s = "AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT"`

Output: `["AAAAACCCCC","CCCCCAAAAA"]`

Example 2:

Input: `s = "AAAAAAAAAAAAA"`

Output: ["AAAAAAAAAA"]

5. There are an infinite amount of bags on a number line, one bag for each coordinate. Some of these bags contain coins. You are given a 2D array coins, where $\text{coins}[i] = [l_i, r_i, c_i]$ denotes that every bag from l_i to r_i contains c_i coins. The segments that coins contain are non-overlapping. You are also given an integer k . Return the maximum amount of coins you can obtain by collecting k consecutive bags.

Example 1:

Input: coins = [[8,10,1],[1,3,2],[5,6,4]], $k = 4$

Output: 10

Explanation:

Selecting bags at positions [3, 4, 5, 6] gives the maximum number of coins: $2 + 0 + 4 + 4 = 10$.

Example 2:

Input: coins = [[1,10,3]], $k = 2$

Output: 6